In [53]:
```
import numpy as np
import pandas as pd
```

concat()

In [56]:
```
# pd.concat(objs, axis = 0, join = "outer", ignore_index = False)
```

In [58]:
```
batch1 = {"RollNo":[101,102,103,104,105],
          "Name" : ["Astha","Ansi","Dvija","Vidhi","Pritam"],
         "Marks" : [25,24,26,22,26]}
```

In [60]:
```
df1 = pd.DataFrame(batch1)
```

In [62]:
```
df1
```

Out[62]:

	RollNo	Name	Marks
0	101	Astha	25
1	102	Ansi	24
2	103	Dvija	26
3	104	Vidhi	22
4	105	Pritam	26

In [64]:
```
batch2 = {"RollNo":[106,107],
          "Name" : ["Bansi","Leena",],
         "Marks" : [25,24]}
```

In [66]:
```
df2 = pd.DataFrame(batch2)
```

In [68]:
```
df2
```

Out[68]:

	RollNo	Name	Marks
0	106	Bansi	25
1	107	Leena	24

In [70]:
```
final = pd.concat([df1,df2], axis = 0, ignore_index=True)
```

In [72]:
```
final.set_index("RollNo", inplace = True)
```

In [74]:
```
final
```

Out[74]:

RollNo	Name	Marks
101	Astha	25
102	Ansi	24
103	Dvija	26
104	Vidhi	22
105	Pritam	26
106	Bansi	25
107	Leena	24

In [76]:
```
df1 = pd.DataFrame({"Name" : ["Jay"], "Age" : [25]})
```

In [78]:
```
df1
```

Out[78]:

	Name	Age
0	Jay	25

In [80]:
```
df2 = pd.DataFrame({"Name" : ["Smita"], "City" : ["Ahmedabad"]})
```

In [82]:
```
df2
```

Out[82]:

	Name	City
0	Smita	Ahmedabad

In [84]:
```
pd.concat([df1,df2], join = "inner")
```

Out[84]:

	Name
0	Jay
0	Smita

In [86]:
```
import numpy as np
import pandas as pd
```

In [88]:
```
df1 = pd.read_csv("students_data.csv")
```

In [90]:
```
df1
```

Out[90]:

	RollNo	Name	Age	Course	City
0	101	Rahul	20	BCA	Ahmedabad
1	102	Priya	21	BBA	Rajkot
2	105	Karan	22	MCA	Vadodara
3	106	Anjali	21	BCA	Ahmedabad

In [98]:
```
df2 = pd.read_csv("student_marks.csv")
```

In [100…
```
df2
```

Out[100…

	RollNo	Name	Marks
0	101	Jay	85
1	102	Neera	91
2	103	Hemal	78
3	104	Riya	88
4	105	Arushi	76

In [102…
```
pd.concat([df1,df2], axis = 1)
```

Out[102…

	RollNo	Name	Age	Course	City	RollNo	Name	Marks
0	101.0	Rahul	20.0	BCA	Ahmedabad	101	Jay	85
1	102.0	Priya	21.0	BBA	Rajkot	102	Neera	91
2	105.0	Karan	22.0	MCA	Vadodara	103	Hemal	78
3	106.0	Anjali	21.0	BCA	Ahmedabad	104	Riya	88
4	NaN	NaN	NaN	NaN	NaN	105	Arushi	76

In [106…
```
pd.merge(df1, df2, on = "RollNo", how = "inner")
```

Out[106…

	RollNo	Name_x	Age	Course	City	Name_y	Marks
0	101	Rahul	20	BCA	Ahmedabad	Jay	85
1	102	Priya	21	BBA	Rajkot	Neera	91
2	105	Karan	22	MCA	Vadodara	Arushi	76

In [108…
```
pd.merge(df1, df2, on = "RollNo", how = "inner", suffixes=("_df1","_df2"))
```

Out[108...

	RollNo	Name_df1	Age	Course	City	Name_df2	Marks
0	101	Rahul	20	BCA	Ahmedabad	Jay	85
1	102	Priya	21	BBA	Rajkot	Neera	91
2	105	Karan	22	MCA	Vadodara	Arushi	76

In [110...
```
## change the column name of df2 to Student_ID
```

In [112...
```
df2 = pd.read_csv("student_marks.csv")
```

In [114...
```
pd.merge(df1, df2, left_on = "RollNo", right_on = "Student_ID", how = "right", s
```

Out[114...

	RollNo	Name_df1	Age	Course	City	Student_ID	Name_df2	Marks
0	101.0	Rahul	20.0	BCA	Ahmedabad	101	Jay	85
1	102.0	Priya	21.0	BBA	Rajkot	102	Neera	91
2	NaN	NaN	NaN	NaN	NaN	103	Hemal	78
3	NaN	NaN	NaN	NaN	NaN	104	Riya	88
4	105.0	Karan	22.0	MCA	Vadodara	105	Arushi	76

In [119...
```
## Again rename to RollNo
```

In [141...
```
df1 = pd.read_csv("students_data.csv")
```

In [143...
```
df2 = pd.read_csv("student_marks.csv")
```

In [145...
```
df1
```

Out[145...

	RollNo	Name	Age	Course	City
0	101	Rahul	20	BCA	Ahmedabad
1	102	Priya	21	BBA	Rajkot
2	105	Karan	22	MCA	Vadodara
3	106	Anjali	21	BCA	Ahmedabad

In [147...
```
df2
```

Out[147...

	RollNo	Name	Marks
0	101	Jay	85
1	102	Neera	91
2	103	Hemal	78
3	104	Riya	88
4	105	Arushi	76

In [157…
```
pd.merge(df1,df2, on =["RollNo","Name"], how="outer")
```

Out[157…

	RollNo	Name	Age	Course	City	Marks
0	101	Jay	NaN	NaN	NaN	85.0
1	101	Rahul	20.0	BCA	Ahmedabad	NaN
2	102	Neera	NaN	NaN	NaN	91.0
3	102	Priya	21.0	BBA	Rajkot	NaN
4	103	Hemal	NaN	NaN	NaN	78.0
5	104	Riya	NaN	NaN	NaN	88.0
6	105	Arushi	NaN	NaN	NaN	76.0
7	105	Karan	22.0	MCA	Vadodara	NaN
8	106	Anjali	21.0	BCA	Ahmedabad	NaN

In [43]:
```
df1
```

Out[43]:

	RollNo	Name	Age	Course	City
0	101	Rahul	20	BCA	Ahmedabad
1	102	Priya	21	BBA	Rajkot
2	105	Karan	22	MCA	Vadodara
3	106	Anjali	21	BCA	Ahmedabad

In [159…
```
df1 = df1.set_index("RollNo")
```

In [161…
```
df1
```

Out[161…

RollNo	Name	Age	Course	City
101	Rahul	20	BCA	Ahmedabad
102	Priya	21	BBA	Rajkot
105	Karan	22	MCA	Vadodara
106	Anjali	21	BCA	Ahmedabad

In [165…
```
df2 = df2.set_index("RollNo")
```

In [167…
```
df2
```

Out[167...

RollNo	Name	Marks
101	Jay	85
102	Neera	91
103	Hemal	78
104	Riya	88
105	Arushi	76

In [169...
```
pd.merge(df1,df2,left_index=True, right_index=True)
```

Out[169...

RollNo	Name_x	Age	Course	City	Name_y	Marks
101	Rahul	20	BCA	Ahmedabad	Jay	85
102	Priya	21	BBA	Rajkot	Neera	91
105	Karan	22	MCA	Vadodara	Arushi	76

In [171...
```
df2 = df2.reset_index()
```

In [175...
```
pd.merge(df1,df2, left_index=True, right_on = "RollNo")
```

Out[175...

	Name_x	Age	Course	City	RollNo	Name_y	Marks
0	Rahul	20	BCA	Ahmedabad	101	Jay	85
1	Priya	21	BBA	Rajkot	102	Neera	91
4	Karan	22	MCA	Vadodara	105	Arushi	76

In [177...
```
df2
```

Out[177...

	RollNo	Name	Marks
0	101	Jay	85
1	102	Neera	91
2	103	Hemal	78
3	104	Riya	88
4	105	Arushi	76

In [179...
```
df1.reset_index(inplace=True)
```

In [181...
```
df1
```

Out[181…

	RollNo	Name	Age	Course	City
0	101	Rahul	20	BCA	Ahmedabad
1	102	Priya	21	BBA	Rajkot
2	105	Karan	22	MCA	Vadodara
3	106	Anjali	21	BCA	Ahmedabad

In [ ]:
```
df1 = df1.set_index("RollNo")
```

In [193…
```
df2 = df2.set_index("RollNo")
```

In [195…
```
df1
```

Out[195…

RollNo	Name	Age	Course	City
101	Rahul	20	BCA	Ahmedabad
102	Priya	21	BBA	Rajkot
105	Karan	22	MCA	Vadodara
106	Anjali	21	BCA	Ahmedabad

In [197…
```
df2
```

Out[197…

RollNo	Name	Marks
101	Jay	85
102	Neera	91
103	Hemal	78
104	Riya	88
105	Arushi	76

In [101…
```
df1
```

Out[101…

RollNo	Name	Age	Course	City
101	Rahul	20	BCA	Ahmedabad
102	Priya	21	BBA	Rajkot
105	Karan	22	MCA	Vadodara
106	Anjali	21	BCA	Ahmedabad

In [203…
```
df1.join(df2, how="left")
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[203], line 1
----> 1 df1.join(df2, how="left")

File ~\anaconda3\Lib\site-packages\pandas\core\frame.py:10764, in DataFrame.join
(self, other, on, how, lsuffix, rsuffix, sort, validate)
  10754     if how == "cross":
  10755         return merge(
  10756             self,
  10757             other,
   (...)
  10762             validate=validate,
  10763         )
> 10764     return merge(
  10765         self,
  10766         other,
  10767         left_on=on,
  10768         how=how,
  10769         left_index=on is None,
  10770         right_index=True,
  10771         suffixes=(lsuffix, rsuffix),
  10772         sort=sort,
  10773         validate=validate,
  10774     )
  10775 else:
  10776     if on is not None:

File ~\anaconda3\Lib\site-packages\pandas\core\reshape\merge.py:184, in merge(lef
t, right, how, on, left_on, right_on, left_index, right_index, sort, suffixes, co
py, indicator, validate)
    169 else:
    170     op = _MergeOperation(
    171         left_df,
    172         right_df,
   (...)
    182         validate=validate,
    183     )
--> 184     return op.get_result(copy=copy)

File ~\anaconda3\Lib\site-packages\pandas\core\reshape\merge.py:888, in _MergeOpe
ration.get_result(self, copy)
    884     self.left, self.right = self._indicator_pre_merge(self.left, self.rig
ht)
    886 join_index, left_indexer, right_indexer = self._get_join_info()
--> 888 result = self._reindex_and_concat(
    889     join_index, left_indexer, right_indexer, copy=copy
    890 )
    891 result = result.__finalize__(self, method=self._merge_type)
    893 if self.indicator:

File ~\anaconda3\Lib\site-packages\pandas\core\reshape\merge.py:840, in _MergeOpe
ration._reindex_and_concat(self, join_index, left_indexer, right_indexer, copy)
    837 left = self.left[:]
    838 right = self.right[:]
--> 840 llabels, rlabels = _items_overlap_with_suffix(
    841     self.left._info_axis, self.right._info_axis, self.suffixes
    842 )
    844 if left_indexer is not None and not is_range_indexer(left_indexer, len(le
ft)):
```

```
    845     # Pinning the index here (and in the right code just below) is not
    846     #  necessary, but makes the `.take` more performant if we have e.g.
    847     #  a MultiIndex for left.index.
    848     lmgr = left._mgr.reindex_indexer(
    849         join_index,
    850         left_indexer,
   (...)
    855         use_na_proxy=True,
    856     )

File ~\anaconda3\Lib\site-packages\pandas\core\reshape\merge.py:2721, in _items_overlap_with_suffix(left, right, suffixes)
   2718 lsuffix, rsuffix = suffixes
   2720 if not lsuffix and not rsuffix:
-> 2721     raise ValueError(f"columns overlap but no suffix specified: {to_rename}")
   2723 def renamer(x, suffix: str | None):
   2724     """
   2725     Rename the left and right indices.
   2726
   (...)
   2737     x : renamed column name
   2738     """

ValueError: columns overlap but no suffix specified: Index(['Name'], dtype='object')
```

In [205…

```
df1.join(df2, how="left", lsuffix="_df1", rsuffix="_df2")
```

Out[205…

RollNo	Name_df1	Age	Course	City	Name_df2	Marks
101	Rahul	20	BCA	Ahmedabad	Jay	85.0
102	Priya	21	BBA	Rajkot	Neera	91.0
105	Karan	22	MCA	Vadodara	Arushi	76.0
106	Anjali	21	BCA	Ahmedabad	NaN	NaN

In [ ]: